

Homework 1 Solutions

Based on Lecture 1: Coordinates, Lines, and Circles.

Exercise 1. Plot $(3, -2)$, $(-4, 1)$, and $(0, 5)$, and say which quadrant (or axis) each lies in. Where does each point go under $(a, b) \mapsto (a + 3, b - 1)$?

Apply $(a, b) \mapsto (a + 3, b - 1)$ to each point:

- $(3, -2)$ is in Quadrant IV, and maps to $(6, -3)$, also in Quadrant IV.
- $(-4, 1)$ is in Quadrant II, and maps to $(-1, 0)$, on the x -axis.
- $(0, 5)$ is on the positive y -axis, and maps to $(3, 4)$, in Quadrant I.

Exercise 2. Find the distance between $(1, 1)$ and $(4, 5)$.

For $(1, 1)$ and $(4, 5)$,

$$d = \sqrt{(4-1)^2 + (5-1)^2} = \sqrt{3^2 + 4^2} = 5.$$

Exercise 3. Show that $(0, 0)$, $(3, 0)$, and $(0, 4)$ form a right triangle using the distance formula.

Let $A = (0, 0)$, $B = (3, 0)$, and $C = (0, 4)$. The three side lengths are

$$AB = 3, \quad AC = 4,$$

and

$$BC = \sqrt{(3-0)^2 + (0-4)^2} = \sqrt{9 + 16} = 5.$$

Since $3^2 + 4^2 = 5^2$, the converse of the Pythagorean theorem shows that the angle at A is a right angle. Thus the points form a right triangle.

Exercise 4. Find the midpoint of $(-3, 4)$ and $(5, -2)$, and verify by the distance formula that it is equidistant from both.

The midpoint is

$$M = \left(\frac{-3+5}{2}, \frac{4+(-2)}{2} \right) = (1, 1).$$

Its distances to the endpoints are

$$\begin{aligned} MA &= \sqrt{(1-(-3))^2 + (1-4)^2} = \sqrt{4^2 + (-3)^2} = 5, \\ MB &= \sqrt{(1-5)^2 + (1-(-2))^2} = \sqrt{(-4)^2 + 3^2} = 5. \end{aligned}$$

Therefore M is equidistant from the two endpoints, as required.

Exercise 5. A point $(x, 0)$ on the x -axis is equidistant from $(1, 2)$ and $(5, -4)$. Find x .

Let $P = (x, 0)$. Equidistance from $(1, 2)$ and $(5, -4)$ gives

$$(x-1)^2 + (0-2)^2 = (x-5)^2 + (0+4)^2.$$

Expanding and simplifying,

$$x^2 - 2x + 5 = x^2 - 10x + 41,$$

so $8x = 36$ and

$$\boxed{x = \frac{9}{2}}.$$

Thus the point is $(\frac{9}{2}, 0)$.

Exercise 6. Write the equation of the circle with center $(2, -1)$ and radius 3.

Using center $(h, k) = (2, -1)$ and radius $r = 3$,

$$\boxed{(x - 2)^2 + (y + 1)^2 = 9}.$$

Exercise 7. Does the point $(4, 3)$ lie on, inside, or outside the circle $x^2 + y^2 = 20$?

The squared distance from $(4, 3)$ to the origin is

$$4^2 + 3^2 = 25.$$

The circle $x^2 + y^2 = 20$ has radius squared 20. Since $25 > 20$, the point $(4, 3)$ lies outside the circle.

Exercise 8. The equation $x^2 + y^2 - 4x + 6y - 12 = 0$ represents a circle. Complete the square to write it in standard form $(x - h)^2 + (y - k)^2 = r^2$, then find its center and radius.

Complete the square:

$$\begin{aligned}x^2 + y^2 - 4x + 6y - 12 &= 0 \\(x^2 - 4x + 4) + (y^2 + 6y + 9) &= 12 + 4 + 9 \\(x - 2)^2 + (y + 3)^2 &= 25.\end{aligned}$$

Therefore the center is

$$\boxed{(2, -3)}$$

and the radius is

$$\boxed{5}.$$

Exercise 9. Find the equation of the circle passing through the three points $(0, 0)$, $(4, 0)$, and $(0, -2)$.

Use the general circle equation

$$x^2 + y^2 + Dx + Ey + F = 0.$$

Substitution of $(0, 0)$ gives $F = 0$. Substitution of $(4, 0)$ gives

$$16 + 4D = 0 \implies D = -4.$$

Substitution of $(0, -2)$ gives

$$4 - 2E = 0 \implies E = 2.$$

Thus the circle is

$$\boxed{x^2 + y^2 - 4x + 2y = 0}.$$

Completing the square gives

$$(x - 2)^2 + (y + 1)^2 = 5.$$

So its center is $(2, -1)$ and its radius is $\sqrt{5}$.

Exercise 10. Compute the slope between $(-1, 3)$ and $(2, -3)$. Is the line rising or falling?

The slope is

$$m = \frac{-3 - 3}{2 - (-1)} = \frac{-6}{3} = -2.$$

Because the slope is negative, the line is falling from left to right.

Exercise 11. The points $(1, 2)$, $(3, k)$, and $(7, 14)$ are collinear. Find k .

The slope from $(1, 2)$ to $(7, 14)$ is

$$\frac{14 - 2}{7 - 1} = \frac{12}{6} = 2.$$

Thus the line through $(1, 2)$ is $y - 2 = 2(x - 1)$, or $y = 2x$. At $x = 3$,

$$\boxed{k = 6}.$$

Exercise 12. Find the equation of the line through $(0, 5)$ and $(2, 1)$.

The slope through $(0, 5)$ and $(2, 1)$ is

$$m = \frac{1 - 5}{2 - 0} = -2.$$

Using $(0, 5)$ in point-slope form,

$$y - 5 = -2(x - 0),$$

so the equation is

$$\boxed{y = -2x + 5}.$$

Exercise 13. A line through $(a, 2)$ and $(3, b)$ has slope 2 and passes through $(0, 1)$. Find a and b .

A line of slope 2 passing through $(0, 1)$ has equation

$$y = 2x + 1.$$

Since $(a, 2)$ lies on it,

$$2 = 2a + 1 \implies a = \frac{1}{2}.$$

Since $(3, b)$ lies on it,

$$b = 2(3) + 1 = 7.$$

Therefore

$$\boxed{a = \frac{1}{2}, \quad b = 7}.$$

Exercise 14. Find the intersection of $y = 2x + 1$ and $y = -x + 4$.

Set the two expressions for y equal:

$$2x + 1 = -x + 4.$$

Then $3x = 3$, so $x = 1$. Substituting gives $y = 3$. The intersection is

$$\boxed{(1, 3)}.$$

Exercise 15. Find the line through $(1, 1)$ perpendicular to $y = 3x - 2$.

The given line has slope 3, so a perpendicular line has slope $-\frac{1}{3}$. Through $(1, 1)$,

$$y - 1 = -\frac{1}{3}(x - 1).$$

Thus

$$\boxed{y = -\frac{1}{3}x + \frac{4}{3}}.$$

Exercise 16. Find the equation of the line through the intersection of $y = 2x + 1$ and $y = -x + 4$ that is perpendicular to $y = \frac{1}{2}x - 3$.

From Exercise 14, the intersection of $y = 2x + 1$ and $y = -x + 4$ is $(1, 3)$. The line $y = \frac{1}{2}x - 3$ has slope $\frac{1}{2}$, so the required perpendicular line has slope -2 . Through $(1, 3)$,

$$y - 3 = -2(x - 1),$$

which simplifies to

$$\boxed{y = -2x + 5}.$$

Exercise 17. Find the centroid of the triangle with vertices $(1, 1)$, $(5, 1)$, and $(3, 7)$.

The centroid is the coordinate-wise average:

$$G = \left(\frac{1+5+3}{3}, \frac{1+1+7}{3} \right) = \boxed{(3, 3)}.$$

Exercise 18. Two vertices of a triangle are $(0, 0)$ and $(4, 2)$, and the centroid is $(2, 3)$. Find the third vertex. Let the third vertex be (x, y) . Since the centroid is $(2, 3)$,

$$\left(\frac{0+4+x}{3}, \frac{0+2+y}{3}\right) = (2, 3).$$

Therefore

$$4+x=6, \quad 2+y=9,$$

so the third vertex is

$$\boxed{(2, 7)}.$$

Exercise 19. Prove that the centroid divides each median in the ratio $2 : 1$ by a purely geometric argument, without coordinates.

Hint. Apply the midsegment theorem to the segment joining the midpoints of two sides.

Let triangle ABC have midpoint M of BC , midpoint N of AC , and midpoint P of AB . Let G be the intersection of medians AM and BN .

In triangle ABC , segment MN joins the midpoints of BC and AC , so by the midsegment theorem $MN \parallel AB$ and $MN = \frac{1}{2}AB$. Similarly, $MP \parallel AC$ and $MP = \frac{1}{2}AC$.

Consider triangle AMN . Since G lies on AM , draw through G a line parallel to MN (and therefore parallel to AB), meeting AN at Q . The midsegment theorem applied in the appropriate similar-triangle configuration gives the same scale factor on the two median directions. Equivalently, triangles GQN and ABN are similar, and the midpoint relations imply that $AG : GM = 2 : 1$.

The same argument applies to each median, so the centroid divides every median in the ratio

$$\boxed{2 : 1}$$

measured from the vertex.

Exercise 20. For a general triangle with vertices (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) , show directly that all three medians pass through

$$\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right),$$

without assuming the $2 : 1$ ratio result.

Hint. Three points P, Q, R are collinear if and only if

$$\frac{y_Q - y_P}{x_Q - x_P} = \frac{y_R - y_P}{x_R - x_P}.$$

Apply this to the vertex (x_1, y_1) , the midpoint $M_1 = \left(\frac{x_2+x_3}{2}, \frac{y_2+y_3}{2}\right)$, and the centroid; then repeat for the other two medians by symmetry.

Let

$$G = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$$

and let M_1 be the midpoint of the side opposite (x_1, y_1) :

$$M_1 = \left(\frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2} \right).$$

The vectors from (x_1, y_1) are

$$G - (x_1, y_1) = \frac{1}{3}((x_2 + x_3 - 2x_1), (y_2 + y_3 - 2y_1)),$$

and

$$M_1 - (x_1, y_1) = \frac{1}{2}((x_2 + x_3 - 2x_1), (y_2 + y_3 - 2y_1)).$$

Thus

$$G - (x_1, y_1) = \frac{2}{3}(M_1 - (x_1, y_1)),$$

so G lies on the median from (x_1, y_1) to M_1 . The formula is symmetric in the three vertices, so the same calculation shows that G lies on the other two medians as well. Hence all three medians pass through G .

A second coordinate proof by direct elimination. Let

$$p = x_2 - x_1, \quad q = x_3 - x_1, \quad r = y_2 - y_1, \quad s = y_3 - y_1.$$

Write the intersection of the two medians as

$$x = x_1 + X, \quad y = y_1 + Y.$$

The midpoint of BC is

$$M_1 = \left(x_1 + \frac{p+q}{2}, y_1 + \frac{r+s}{2} \right).$$

Using the cross-multiplied form of the two-point formula for the median from $A = (x_1, y_1)$,

$$(p+q)Y - (r+s)X = 0. \tag{1}$$

The midpoint of AC is

$$M_2 = \left(x_1 + \frac{q}{2}, y_1 + \frac{s}{2} \right),$$

while $B = (x_1 + p, y_1 + r)$. The cross-multiplied two-point formula for the median from B is

$$(q-2p)(Y-r) - (s-2r)(X-p) = 0.$$

Expand the constant terms:

$$\begin{aligned} (q-2p)Y - (s-2r)X &= (q-2p)r - (s-2r)p \\ &= qr - 2pr - ps + 2pr \\ &= qr - ps. \end{aligned}$$

Therefore the second median is

$$(q - 2p)Y - (s - 2r)X = qr - ps. \quad (2)$$

Now eliminate Y . Multiply (1) by $(q - 2p)$ and (2) by $(p + q)$, then subtract the second result from the first:

$$\begin{aligned} & ((q - 2p)(p + q)Y - (q - 2p)(r + s)X) \\ & - ((p + q)(q - 2p)Y - (p + q)(s - 2r)X) \\ & = (p + q)(ps - qr). \end{aligned}$$

The Y terms cancel. The coefficient of X is

$$\begin{aligned} & - (q - 2p)(r + s) + (p + q)(s - 2r) \\ & = -qr - qs + 2pr + 2ps + ps + qs - 2pr - 2qr \\ & = 3ps - 3qr = 3(ps - qr). \end{aligned}$$

Thus

$$3(ps - qr)X = (p + q)(ps - qr).$$

For a nondegenerate triangle, $ps - qr \neq 0$, so

$$X = \frac{p + q}{3}.$$

Now eliminate X . Multiply (1) by $(s - 2r)$ and (2) by $(r + s)$, then subtract the second result from the first:

$$\begin{aligned} & ((s - 2r)(p + q)Y - (s - 2r)(r + s)X) \\ & - ((r + s)(q - 2p)Y - (r + s)(s - 2r)X) \\ & = (r + s)(ps - qr). \end{aligned}$$

The X terms cancel. The coefficient of Y is

$$\begin{aligned} & (s - 2r)(p + q) - (r + s)(q - 2p) \\ & = ps + qs - 2pr - 2qr - qr - qs + 2pr + 2ps \\ & = 3ps - 3qr = 3(ps - qr). \end{aligned}$$

Thus

$$3(ps - qr)Y = (r + s)(ps - qr),$$

and, since $ps - qr \neq 0$,

$$Y = \frac{r + s}{3}.$$

Finally,

$$x = x_1 + X = x_1 + \frac{(x_2 - x_1) + (x_3 - x_1)}{3} = \frac{x_1 + x_2 + x_3}{3},$$

$$y = y_1 + Y = y_1 + \frac{(y_2 - y_1) + (y_3 - y_1)}{3} = \frac{y_1 + y_2 + y_3}{3}.$$

Thus the direct coordinate intersection of the first two medians is the centroid.

Exercise 21. Let $A = (0, 0)$, $B = (6, 0)$, and $C = (2, 8)$. Find the equations of the three altitudes of triangle ABC , and show that they meet at a single point. What is that point?

Hint. The side AB is horizontal, so the altitude from C is vertical: write it as $x = \text{constant}$ rather than trying to use a slope.

Let $A = (0, 0)$, $B = (6, 0)$, and $C = (2, 8)$.

The side AB is horizontal, so the altitude from C is vertical:

$$\ell_C : \quad x = 2.$$

The slope of BC is

$$\frac{8 - 0}{2 - 6} = -2,$$

so the altitude from A has slope $\frac{1}{2}$:

$$\ell_A : \quad y = \frac{1}{2}x.$$

The slope of AC is $\frac{8}{2} = 4$, so the altitude from B has slope $-\frac{1}{4}$. Therefore

$$\ell_B : \quad y = -\frac{1}{4}(x - 6) = -\frac{1}{4}x + \frac{3}{2}.$$

At $x = 2$, the first altitude gives $y = 1$, and the second gives

$$-\frac{1}{4}(2) + \frac{3}{2} = 1.$$

Thus all three altitudes meet at

$$\boxed{H = (2, 1)}.$$

Exercise 22. Let $A = (x_1, y_1)$, $B = (x_2, y_2)$, $C = (x_3, y_3)$.

(a) Using the remark at the end of Section 7, show that (x, y) lies on the altitude from A if and only if

$$(x - x_1)(x_3 - x_2) + (y - y_1)(y_3 - y_2) = 0,$$

and write the two analogous equations for the altitudes from B and from C , keeping the same cyclic pattern $A \rightarrow B \rightarrow C \rightarrow A$ of the indices.

(b) Expand all three equations and add them. Show that the sum is identically zero: every term cancels, for every point (x, y) .

(c) Conclude that the three altitudes are concurrent: any point lying on two of them automatically lies on the third. That point is the orthocenter H .

(a)

The side BC has direction vector

$$C - B = (x_3 - x_2, y_3 - y_2).$$

The altitude from A has direction vector $(x - x_1, y - y_1)$, so perpendicularity gives

$$\boxed{(x - x_1)(x_3 - x_2) + (y - y_1)(y_3 - y_2) = 0}.$$

Following the same cyclic pattern, the altitude from B is

$$\boxed{(x - x_2)(x_1 - x_3) + (y - y_2)(y_1 - y_3) = 0},$$

and the altitude from C is

$$\boxed{(x - x_3)(x_2 - x_1) + (y - y_3)(y_2 - y_1) = 0}.$$

(b)

Add the three left-hand sides. The coefficients of x and y are

$$(x_3 - x_2) + (x_1 - x_3) + (x_2 - x_1) = 0,$$

and

$$(y_3 - y_2) + (y_1 - y_3) + (y_2 - y_1) = 0.$$

The constant terms also cancel:

$$\begin{aligned} & -x_1(x_3 - x_2) - x_2(x_1 - x_3) - x_3(x_2 - x_1) \\ & = -x_1x_3 + x_1x_2 - x_2x_1 + x_2x_3 - x_3x_2 + x_3x_1 = 0, \end{aligned}$$

and the corresponding y terms cancel in exactly the same way. Therefore the sum of the three equations is identically zero for every (x, y) .

(c)

If a point (x, y) lies on the first two altitudes, the first two left-hand sides are zero. Since the sum of all three left-hand sides is identically zero, the third left-hand side must also be zero. Therefore the point lies on the third altitude.

Hence the three altitudes are concurrent. Their common point is the orthocenter

$$\boxed{H}.$$